Electrical and Servicing Systems

Eduardo Manuel Godinho Rodrigues

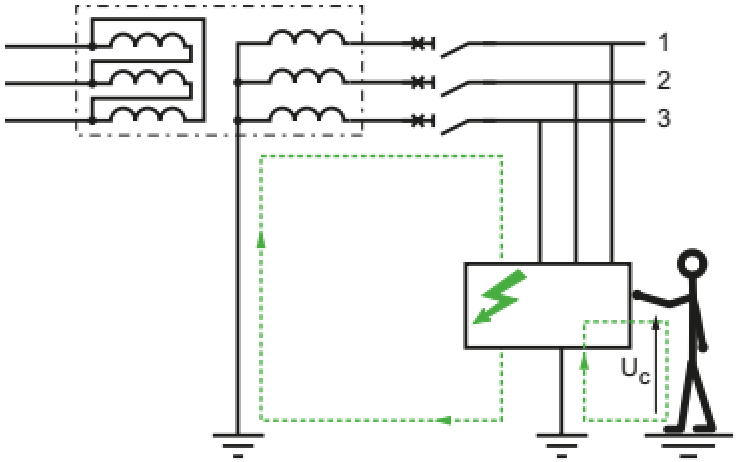
上海大学里斯本学院
ULISBOA SCHOOL, SHANGHAI UNIVERSITY
ULisboa School, Shanghai University

Electrical and Servicing Systems

Protection and Safety Systems

The primary concern associated with electrical safety is the protection from electrical shock hazards that could occur in various fault conditions, which may inflict a threat to life or severe damage to equipment, installations, and facilities. Such hazards are associated with high voltage/high current.

When a power conductor is inadvertently short-circuited to the equipment's enclosure, the AC potential rises to that of the power conductor.



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Dangers to persons due to the contact with a live part are caused by the current flow through the human body. These effects are:

- tetanization

the muscles affected by the current flow involuntary contract and the let-go of conductive parts gripped is difficult. Note: very high currents do not usually induce muscular tetanization because, when the body gets in touch with such currents, the muscular contraction is so sustained that the involuntary muscle movements generally throw the subject away from the conductive part;

- breathing arrest

if the current flows through the muscles controlling the respiratory system, the involuntary contraction of these muscles alters the normal respiratory process and the subject may die due to suffocation or suffer the consequences of traumas caused by asphyxia;

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Dangers to persons due to the contact with a live part are caused by the current flow through the human body. These effects are:

- ventricular fibrillation:

the most dangerous effect is due to the superposition of the external currents with the physiological ones which, by generating uncontrolled contractions, induce alterations of the cardiac cycle. This anomaly may become an irreversible phenomenon since it persists even when the stimulus has ceased;

- burns:

they are due to the heating deriving, by Joule effect, from the current passing through the human body.

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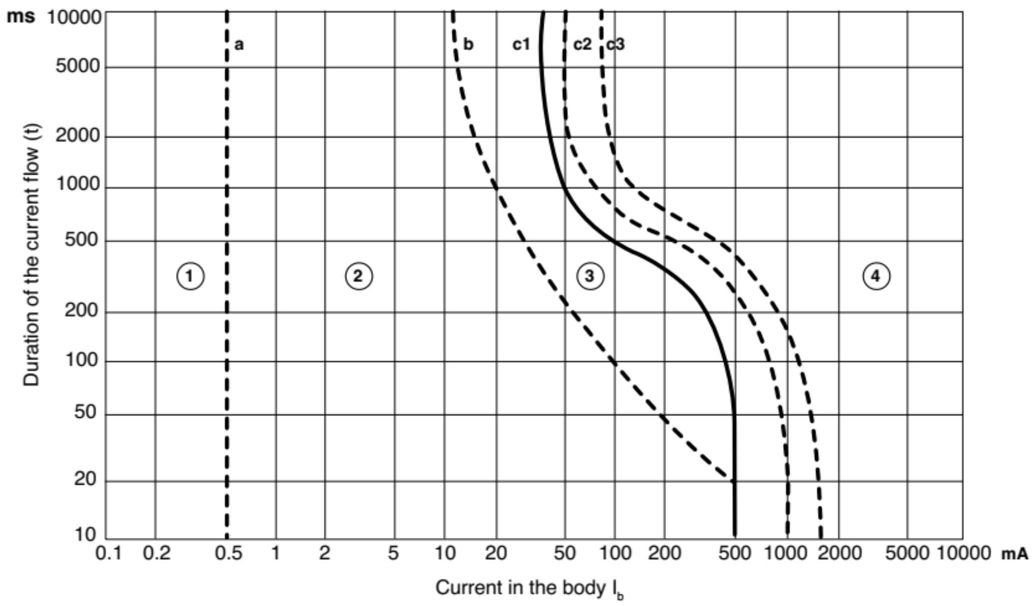
Time-current zones of the effects of alternating current on the human body



Effects of alternating current on human body

Zone	Effects
1	usually no reaction
2	usually no harmful physiological effects
3	usually no organic damage to be expected. Likelihood of cramp like muscular cotractions and difficulty in breathing; reversible disturbances of formation and conduction of impulses in the heart, including atrial fibrillation and transient cardiac arrest without ventricular fibrillation increasing with current magnitude and time
4	in addition to the effects of zone 3, the probability of ventricular fibrillation increases up to about 5% (curve c2), 50% (curve c3) and above 50% beyond the curve c3. Pathophysiological effects such as cardiac arrest, breathing arrest and severe burns may occur increasing with current magnitude and time

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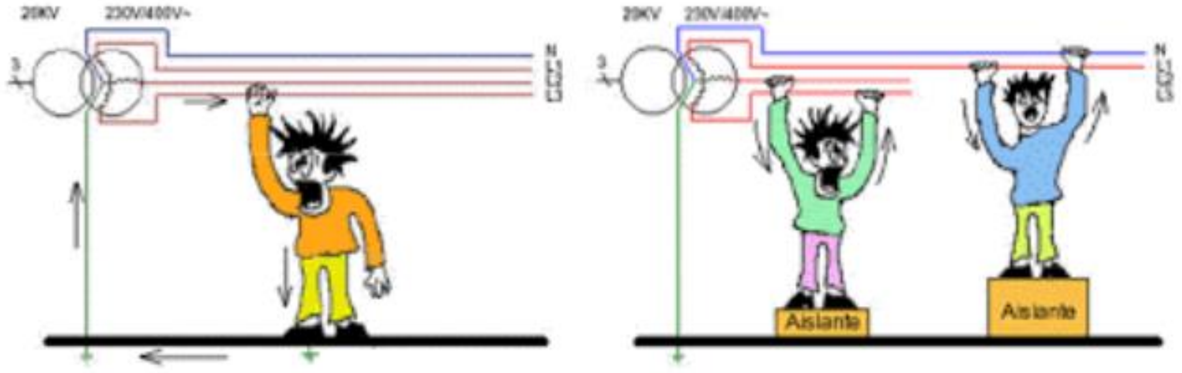
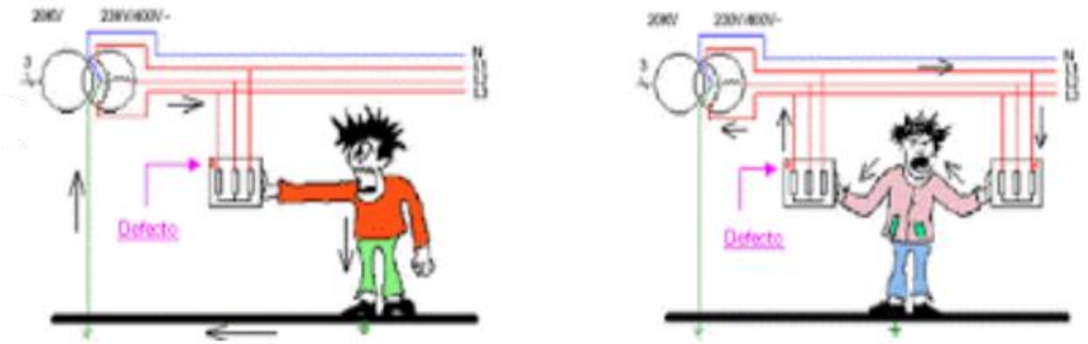
Protective measures against electric shocks are based on two type of events:

Direct contact: this is accidental contact of persons with a live conductor (phase or neutral) or a normally live conductive element.



Indirect contact: contact of a person with accidentally energized metal frames. This accidental energizing is the result of an insulation fault.



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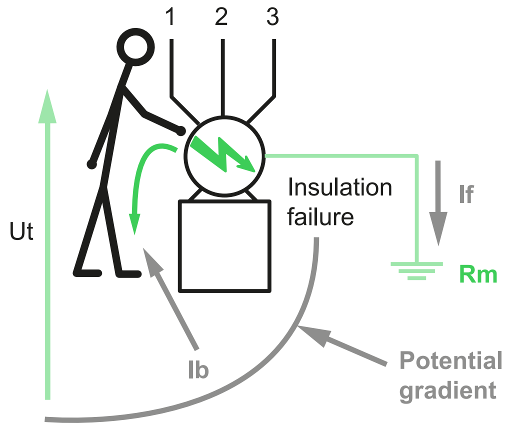
Indirect contact

A fault current flows and creates a potential rise between the frame and the earth, thus causing a fault voltage to appear which is dangerous if it exceeds voltage UL.



The touch voltage Ut is the voltage existing (as the result of insulation failure) between an exposed-conductive-part and any conductive element within reach which is at a different (generally earth) potential.

Current passing through the human body depends on:

- The level of the touch voltage generated by the fault current injected in the earth electrode;
- The resistance of the human body;
- The value of additional resistances like shoes.

Ut : Touch voltage. $Ut \leq Ue$
Ib: Current through the human body.
$Ib = Ut / Rb$
Rb: Resistance of the human body
If: Earth Fault current
Rm: Resistance of the earth electrode

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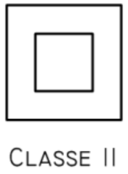

Direct contacts

In LV installations (230/400 V), protection measures consist in placing these live parts out of reach or in insulating them by means of insulators, enclosures or barriers.
A complementary measure against direct contacts consists in using instantaneous $I \leq 30$ mA high sensitivity residual current devices known RCDs.

Plastic cover as barrier

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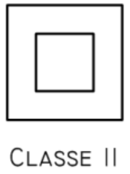
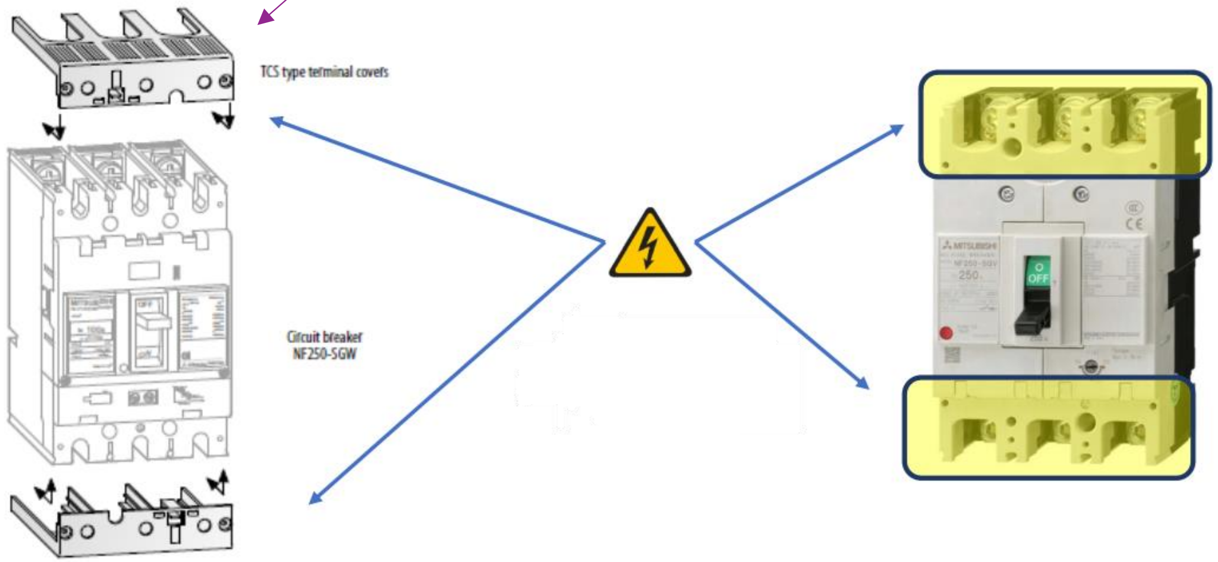


Direct contacts

Enclosure protection for the active parts

Plastic covers that prevent to touch by hand where the conductors are connected.

Circuit breaker



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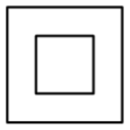


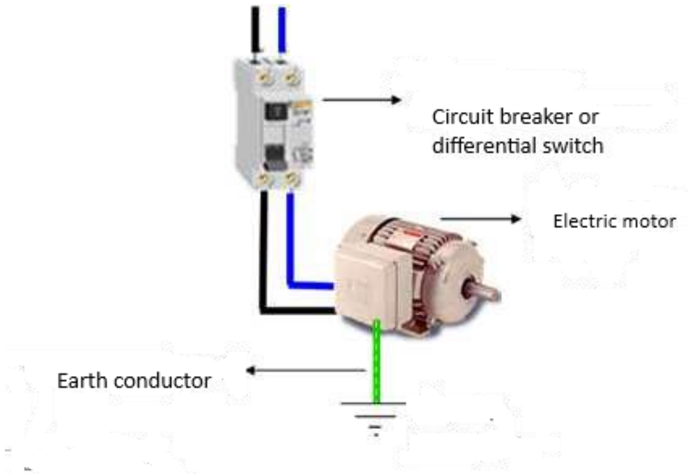
Indirect contacts

Measures to prevent the occurrence of dangerous touch voltages

Use of insulation class II, or automatic interruption of the power supply before the touch voltage becomes dangerous:

- Connection of accessible conductive parts to the protection circuit.
- - Protection by automatic interruption of the power supply (e.g.earth leakage protection).

It is an electrical switchboard in which protection against electric shock is not guaranteed by the main insulation alone. For these class II switchboards additional safety measures are provided, such as double insulation or reinforced insulation.



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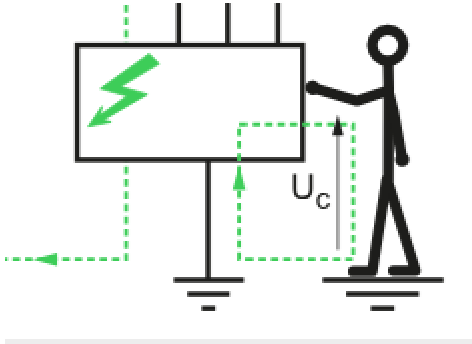
Indirect contacts

IEC 60364-4-41 says:

There must be a protective device that automatically disconnects the circuit or equipment from the power supply when a fault arises between an active part and a ground. This measure of protection against indirect contacts is intended to prevent contact voltages presumed to exceed the following conventional limit voltages (UL) from being maintained between simultaneously accessible conductive parts for a time sufficient to create a risk of dangerous physiopathological effects for persons.



Uo (V AC)		50 < Uo ≤ 120	120 < Uo ≤ 230	230 < Uo ≤ 400	Uo > 400
System	TN	0.8	0.4	0.2	0.1
	TT	0.3	0.2	0.07	0.04

The higher the value of Uc, the higher the rapidity of supply disconnection required to provide protection .The highest value of Uc that can be tolerated indefinitely without danger to human beings is 50 V AC

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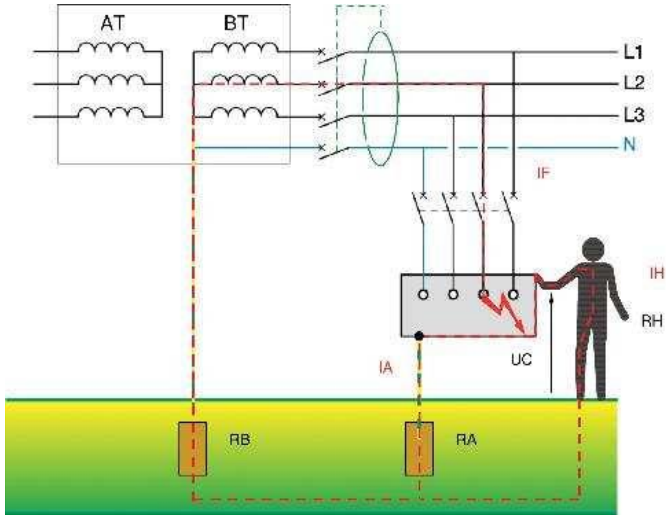
Indirect contacts

Measures to prevent the occurrence of dangerous touch voltages

Automatic disconnection of the power supply before the contact voltage becomes dangerous:
- Connection of accessible conductive parts to the protection circuit;
- Protection by automatic power supply cut-off.



This measure requires coordination between

- the type of power supply system (TT, TN or IT), the impedance of the power supply and the earth system;
- the impedance values of the different line elements and the leakage current circulating through the earthing circuit;
- the characteristics of the protection devices that detect insulation faults.

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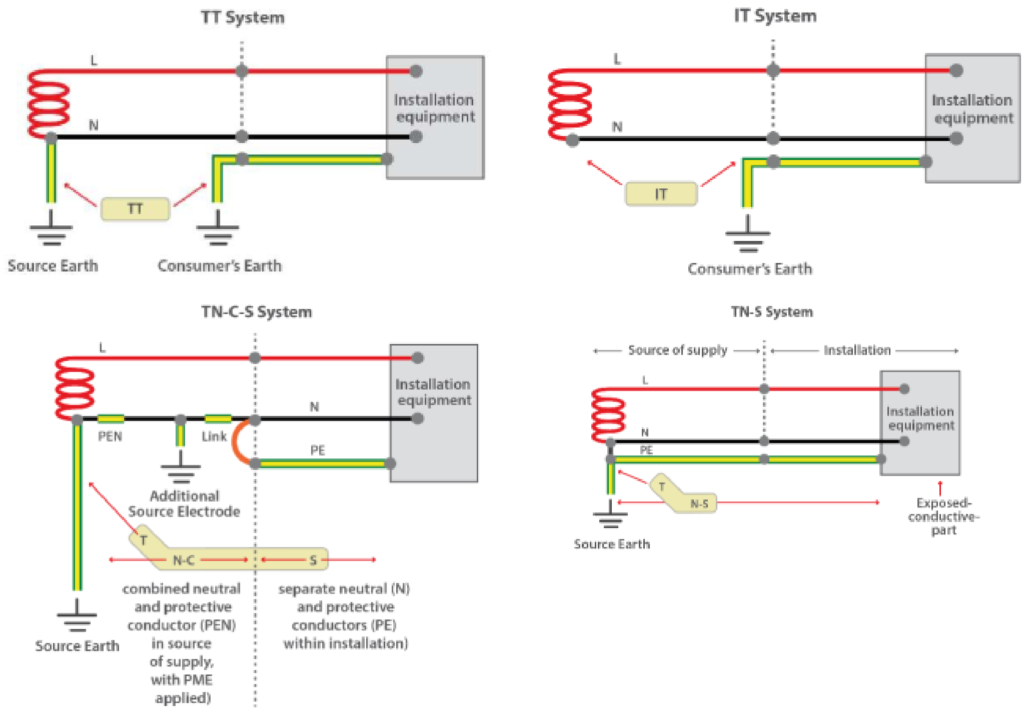
Earthing Systems

The international standard IEC 60364-1 (low voltage installations)installations) distinguishes between three earthing systems, using the letter code TN, TT and IT.

The first letter indicates the type of connection for the Neutral of the transformer (the generator):- ‘T’ direct connection to an earth point- ‘I’ has no connection to the general earth (isolation),- or through a high impedance.

The second letter indicates the connection of the metal masses of the electrical equipment:- ‘T’ direct connection to an earth point- ‘N’ connection to the Neutral of the original installation, which is connected to its own earth.



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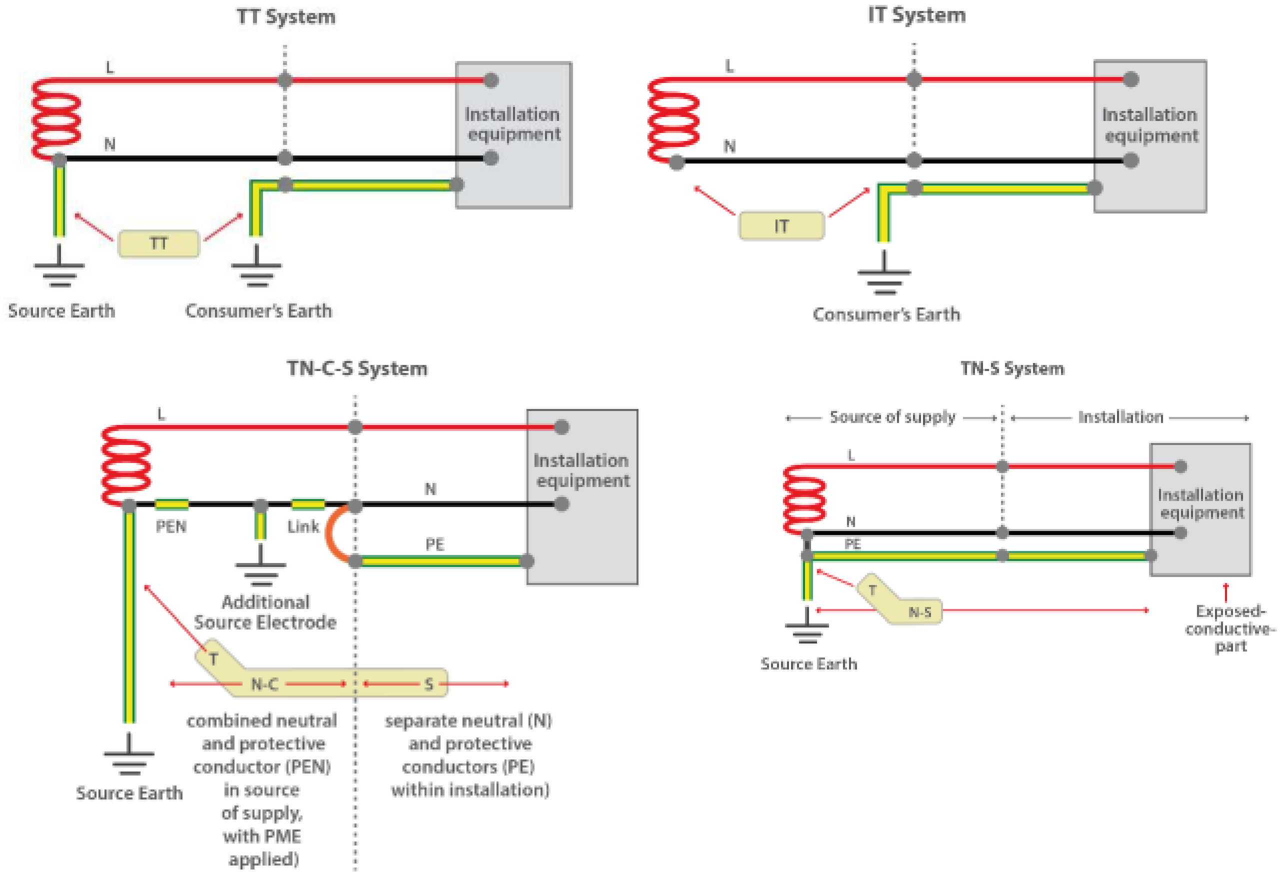
Earthing Systems



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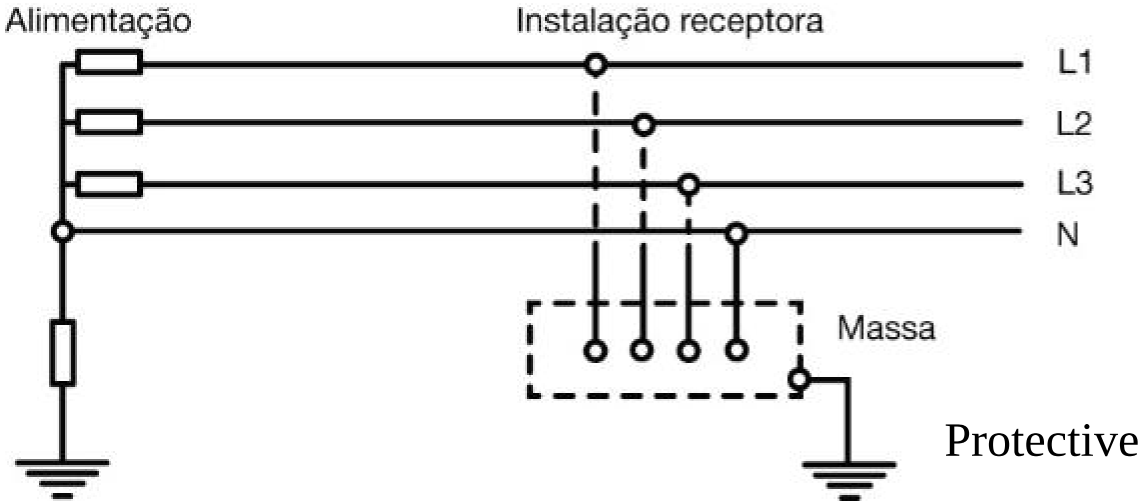
Earthing systems - TT system

The TT neutral regime is characterized by the PT (Transformer Substation) transformer neutral being directly connected to the service earth (RB) and the ground being connected to the protective earth (RA).



Service earth RB

Protective earth (RA)

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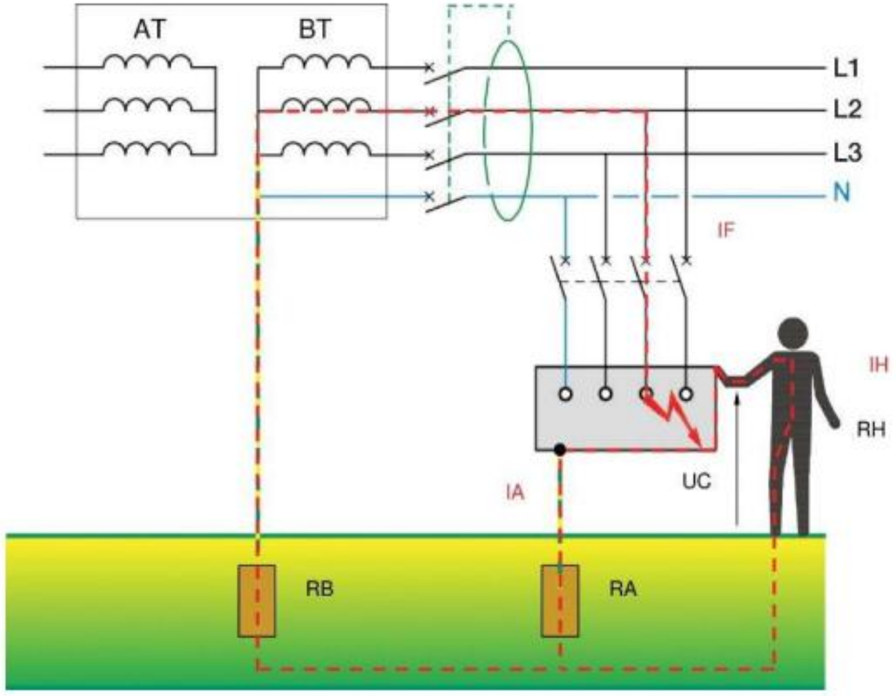
Earthing systems - TT system

Num regime TT o valor máximo admissível da Resistência de terra num PT?

20 Ω



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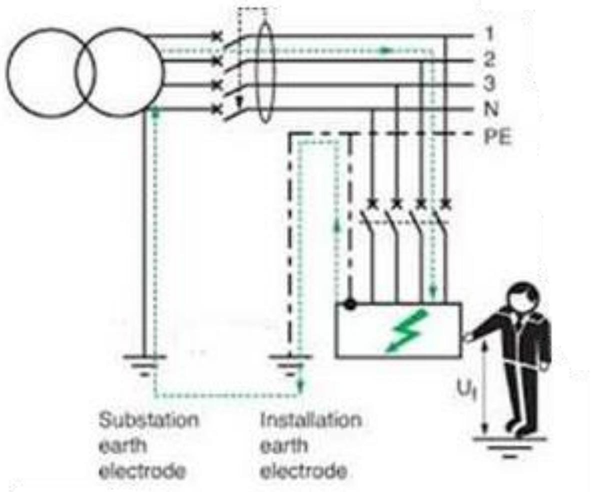
Earthing systems - TT system



Characteristics of the TT system

- The centre point of the transformer is connected to earth;

- All the conductive parts of the equipment are connected to an earth electrode of the same equipment;

- The Neutral conductor must be insulated to prevent a difference in potential (voltage) between N and PE and consequently the circulation of current (I);

- It is advisable to check for earth faults or residual currents (leakage currents).

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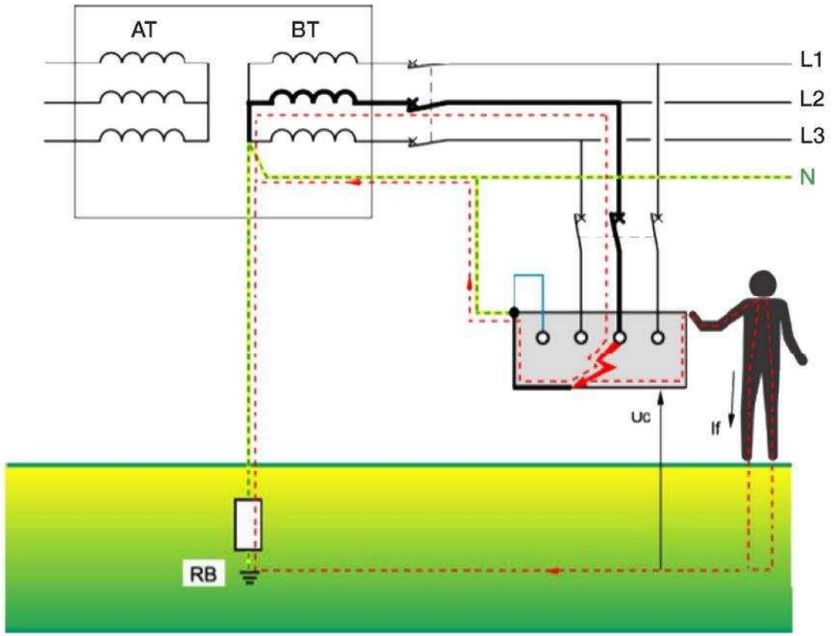
Earthing systems - TN system

The neutral of the PT's transformer is directly connected to the service earth and the ground is directly connected to the neutral via its own conductor (PEN).

PEN conductor (PE + N): Conductor connected to earth and which has protective conductor (PE) and neutral conductor (N) at the same time. and neutral conductor (N).

It is green/yellow throughout its length and at the ends of its connections it must be marked blue.



In a TN regime, what is the maximum permissible value of the earth resistance?

1 Ω

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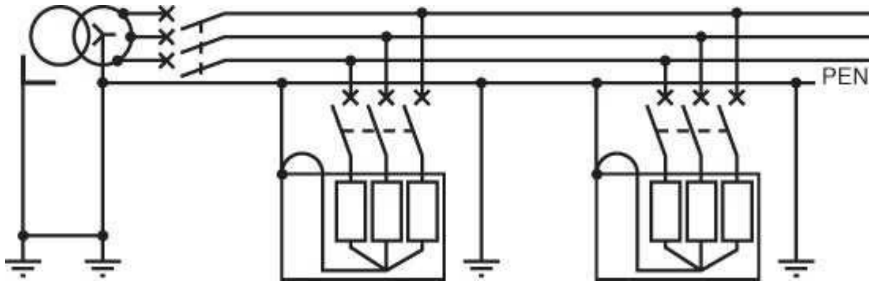
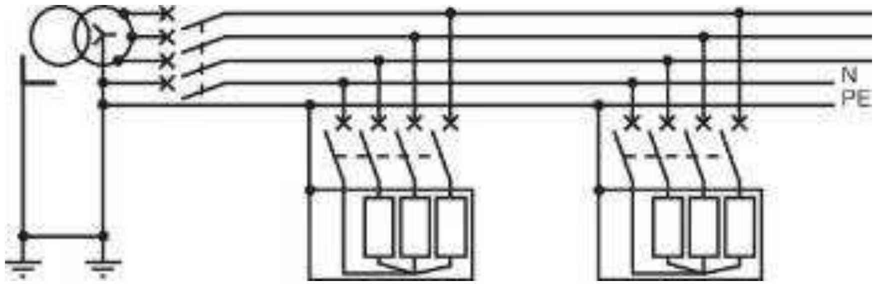
Earthing systems - TN system

It can have 2 different connection modes:

TN-C: The N and PE conductors of the electrical equipment are connected directly to the PEN (NEUTRAL) conductor of the transformer/generator. The PEN conductor must not be insulated or disconnected.



TN-S: The N and PE conductors of the electrical equipment are connected separately in the electrical equipment. Neutral can be disconnected or insulated.



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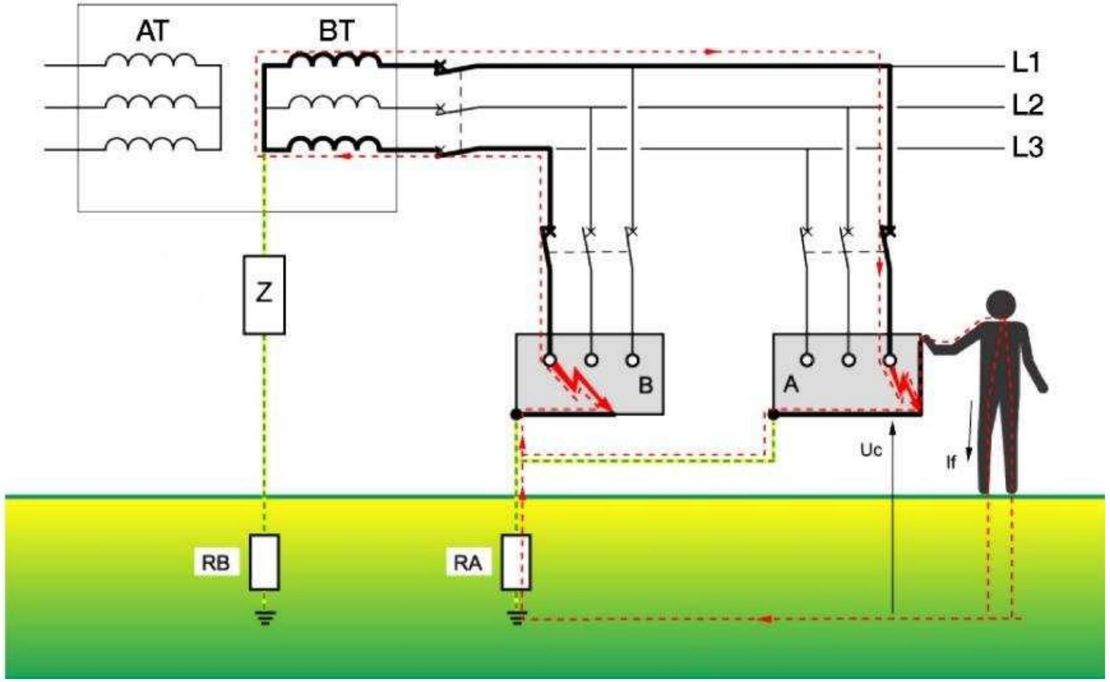
Earthing systems - IT system

The Neutral of the PT transformer is isolated or connected via a service ground impedance (RB)and the ground is directly connected toto the protective earth (RA).



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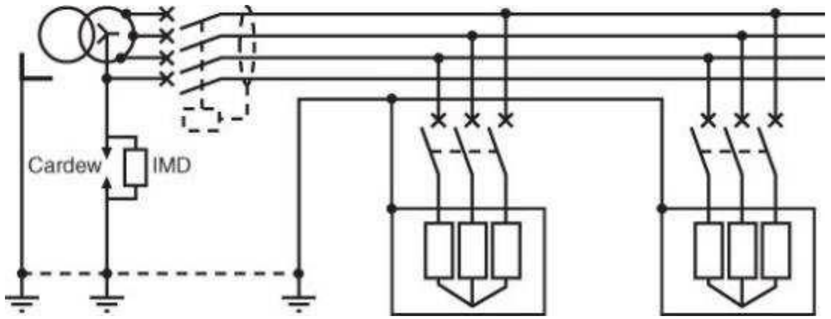

Earthing systems - IT system

Characteristics of the IT system:

- The Neutral of the PT transformer is isolated or connected via an impedance to earth. Neutral is not normally distributed to the installation;

- In the event of a first fault, there must be an insulation meter (CPI) which constantly monitors the installation and raises an alarm in the event of an insulation fault;

- This system is used in applications where the continuity of the installation's service is required (e.g. hospitals) and has a high capacity.(e.g. hospitals) and has high EMC immunity (it may be advisable to use it with high-powered servos or high-power drives).

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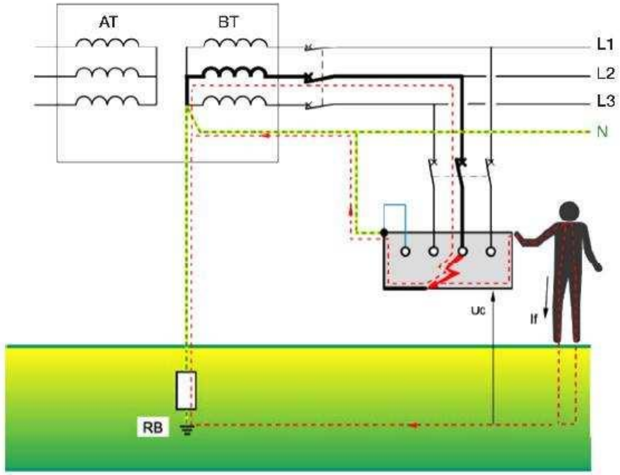
Protection against electric shock - Indirect Contacts

Automatic power cut-off before the contact voltage is dangerous

In the TN regime

Connection of accessible conductive parts to the protection circuit.

Use of overcurrent protective devices (TN-C) or residual current protective devices (RCDs) equipped with overcurrent protective devices (TN-S).



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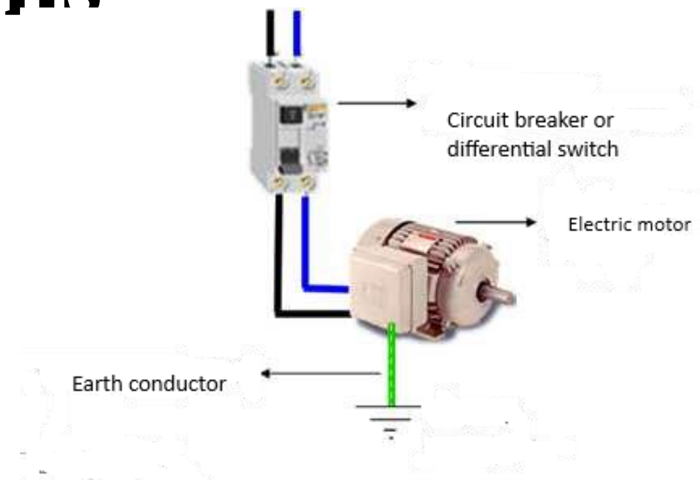
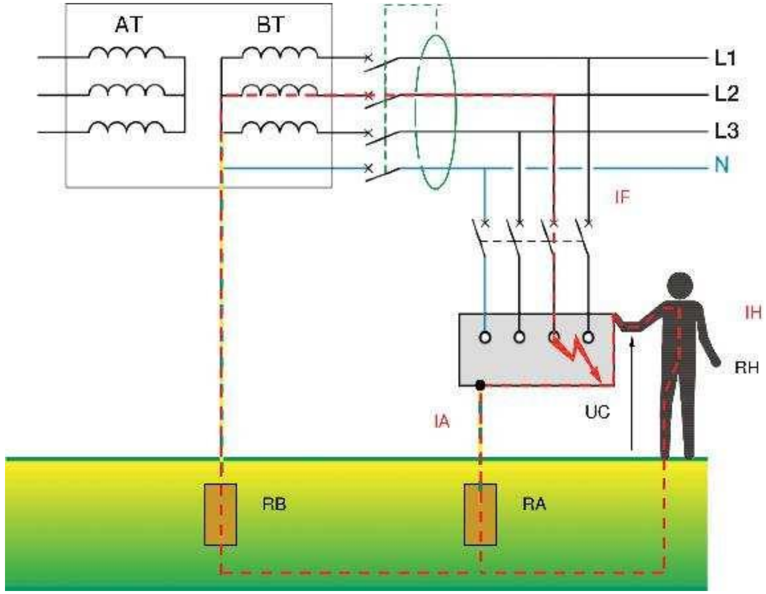
Protection against electric shock - Indirect Contacts

Automatic power cut-off before the contact voltage is dangerous



In the TT regime

Connection of accessible conductive parts to the protection circuit and use of RCDs and overcurrent protection devices, provided that the impedance value of the fault loop Zs is permanently low and reliable.



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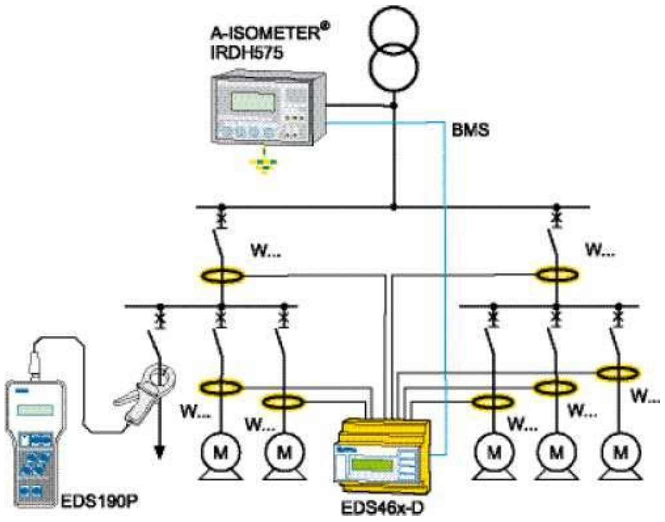
Protection against electric shock - Indirect Contacts

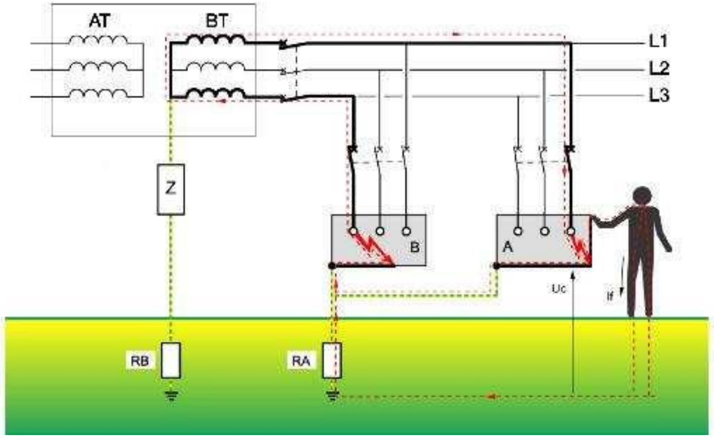
Automatic power cut-off before the contact voltage is dangerous

In the IT regime

Connection of accessible conductive parts to the and during an insulation fault, an acoustic and optical signal must be given and maintained an acoustic and optical signal.

After activation, the acoustic signal can be muted manually. There may be an agreement between the energy supplier and the user regarding the provision of insulation monitoring devices and/or insulation fault localization systems.





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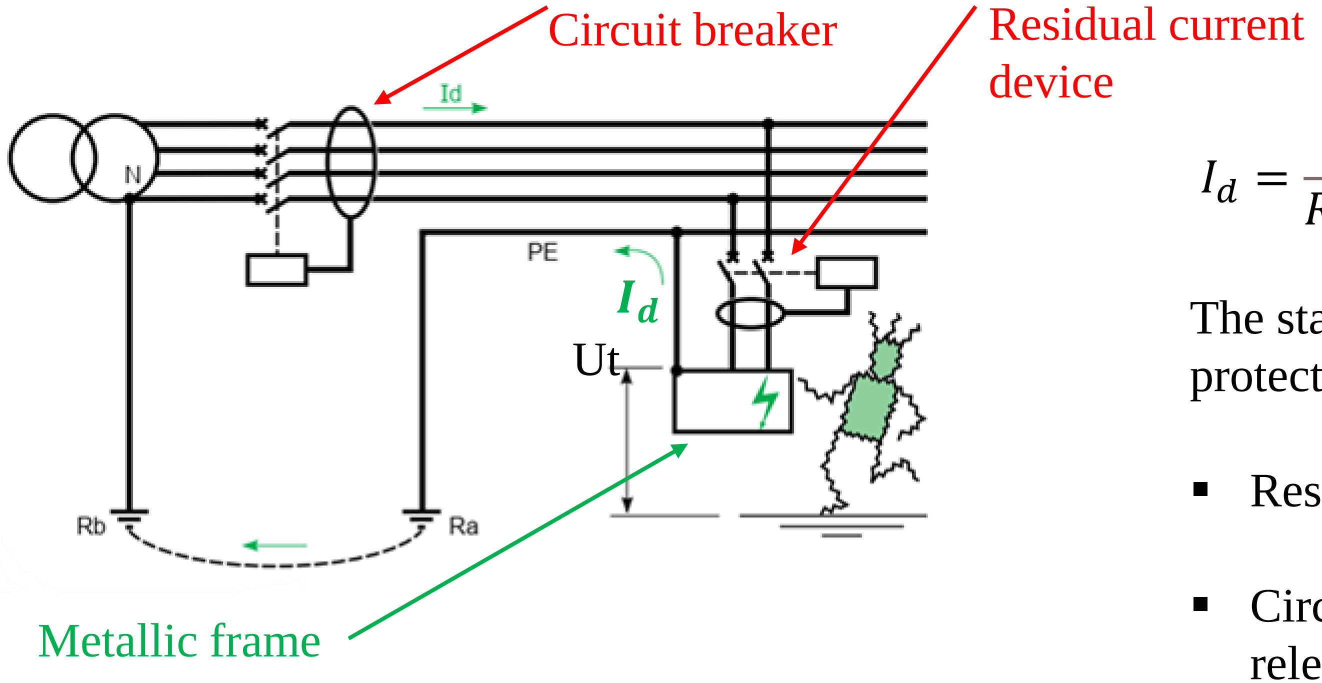
Indirect contact protection in a TT earthing system



The impedance of this fault loop is made up of the resistances of the earth electrodes R_A and R_B, which limits the value of the fault current.

$$I_d = \frac{U_p}{R_A + R_B} \Rightarrow U_t = \frac{R_A}{R_A + R_B} U_p$$

The standards allow two types of automatic protection devices:

- Residual current devices (recommended)
- Circuit breaker with thermomagnetic releases (*)

* Use only when the earth electrode resistances have very low values (a solution that has little applicability)

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Indirect contact protection in a TT earthing system

What is the working principle of a residual current device?

In a healthy circuit, neutral conductor current is equal to the current of the phase conductor but of opposite sign. → $I_N \approx I_L$

If the exposed conductive part becomes live due to an insulation failure in one conductor, an additional current will circulate from that conductor to the earth electrode of the installation. → $|I_P - I_N| = I_{res}$

RCDs are designed to trip for residual current values that are very small compared to the normal load currents flowing in the main conductors (mA compared to Amps).

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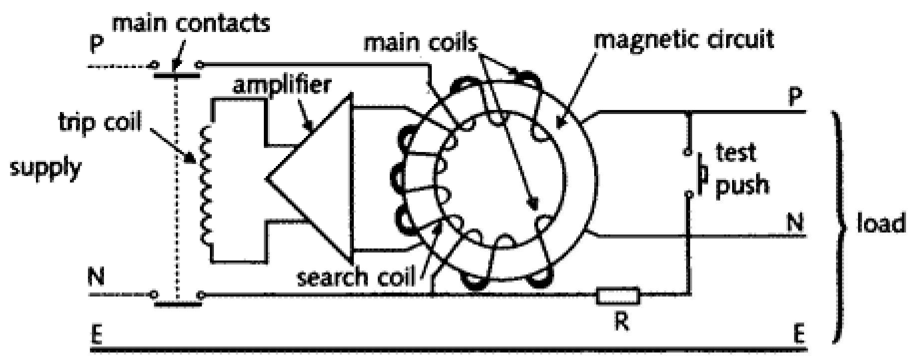
Indirect contact protection in a TT earthing system

What is the working principle of a residual current device?



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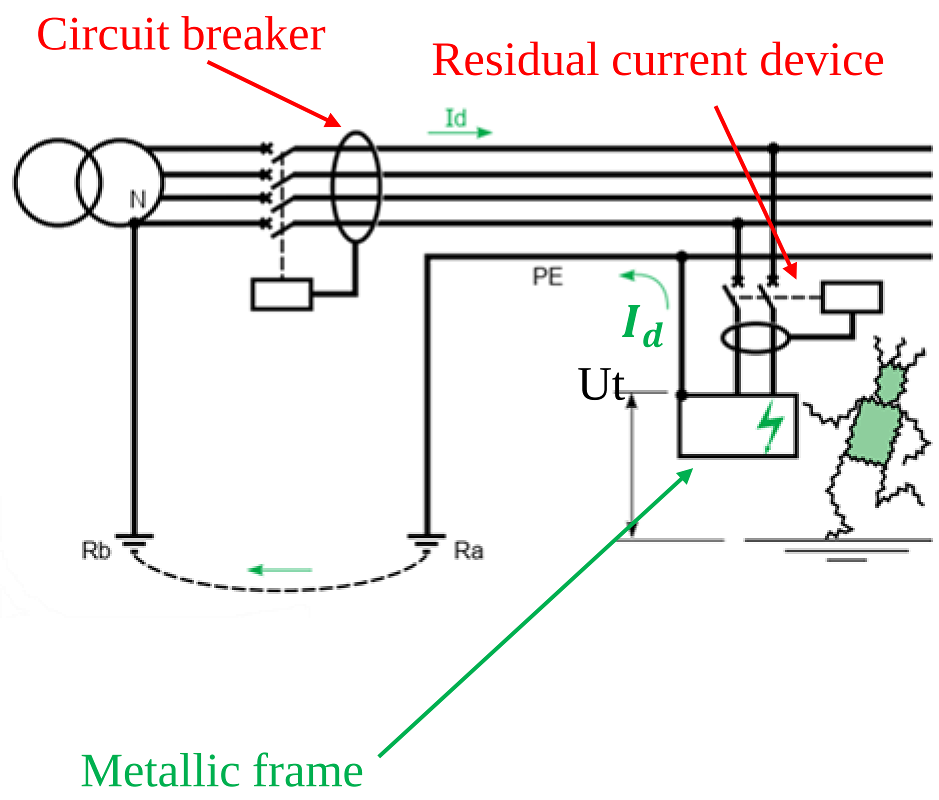
Indirect contact protection in a TT earthing system



The sensitivity ($I_{\Delta n}$) of the residual current device must fulfil the following condition: $I_{\Delta n} = \frac{U_L}{R_A}$

U_L
- 50 V (for most applications)
- 25 V (for temporary supplies (to work sites, …) and agricultural and horticultural places)

The sensitivity value of the residual current device (RCD) depends only on the type of location (UL) and the resistance of the earth electrode that is connected to metallic frame of the equipments (RA).

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Indirect contact protection in a TT earthing system

The procedure for selecting the nominal value of the residual current device must fulfil the following conditions:

$$U_t \leq U_L \qquad I_{\Delta n} \leq I_d \qquad I_{\Delta n} = \frac{U_L}{R_A}$$

$I_{\Delta n}$	Maximum resistance of the earth electrode	
	(50 V)	**(25 V)**
3 A	16 Ω	8 Ω
1 A	50 Ω	25 Ω
500 mA	100 Ω	50 Ω
300 mA	166 Ω	83 Ω
30 mA	1666 Ω	833 Ω

The IEC 60364-4-41 standard specifies the maximum time for protective devices against indirect in the TT scheme. It specifies that for rated current does not exceed 32 A, the fault clearing time must not exceed 0.2 s. On all other circuits, the time must not exceed 1s, and selectivity must be ensured between RCDs on the same distribution circuit.

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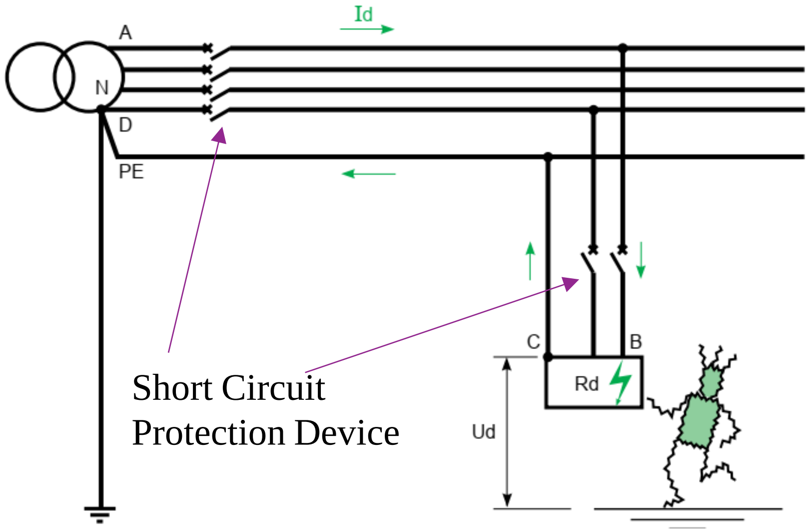
Indirect contact protection in an TN earthing system



When an insulating fault is present, the fault current *Id* is only limited by the impedance of the fault loop cables.

$$I_d = \frac{U_0}{R_{ph1} + R_{PE}}$$

When a short-circuit occurs, it is accepted that the impedances upstream from the relevant feeder cause a voltage drop of around 20 % on phase-to-neutral voltage *Uo*, which is the nominal voltage between phase and earth.

$$U_d = R_{PE} * I_d \Rightarrow U_d = 0.8U_o \frac{R_{PE}}{R_{ph1} + R_{PE}}$$

Typically, the insulation fault originates a short-circuit current event.

For 230/400 V networks, this voltage of around Uo/2 (if R_{PE} = Rph) is dangerous since it exceeds the limit safety voltage (50 V).

$$U_d = U_f$$

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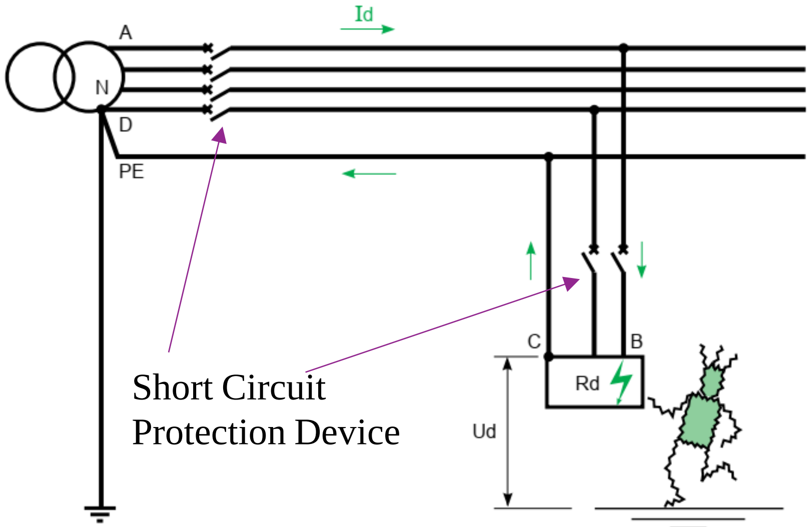
Indirect contact protection in an TN earthing system



As the insulation fault resembles a phase-neutral short-circuit, breaking is achieved by the Short Circuit Protection Device (SCPD) with a maximum specified breaking time depending on conventional contact limit voltage (UL).

To be sure that the protection device really is activated, the current Id must be greater than the operating threshold of the protection device *Ia (Id > Ia)* irrespective of where the fault occurs.

$$I_d = \frac{0.8U_o}{Z} = \frac{0.8U_o}{R_{ph} + R_{PE}} = \frac{0.8U_o S_{ph}}{\rho(1+m)L}$$

$$m = \frac{S_{ph}}{S_{PE}}$$

A method of checking that the protection device will ensure protection of persons is to calculate the maximum length (Lmax) not to be exceeded by each feeder for a given protection threshold Ia.

$$U_d = U_f$$

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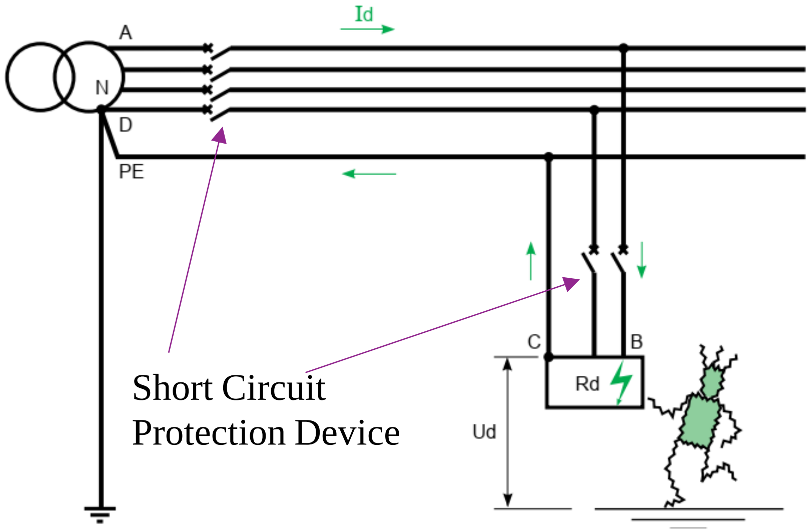
Indirect contact protection in an TN earthing system



Protection device to perform its function properly, Ia must be less than Id, hence the expression of Lmax, the maximum length authorized by the protection device with a threshold Ia.

$$L_{max} = \frac{0.8 U_o S_{ph}}{\rho(1+m) I_a}$$

Lmax: maximum length in m

Uo: phase-to-neutral voltage 230 V for a three phase 400 V network

ρ: resistivity to normal operating temperature

I_a: Automatic breaking current

For a circuit-breaker Ia = Im (Im operating current of the magnetic or short time delay trip release)

$$U_d = U_f$$

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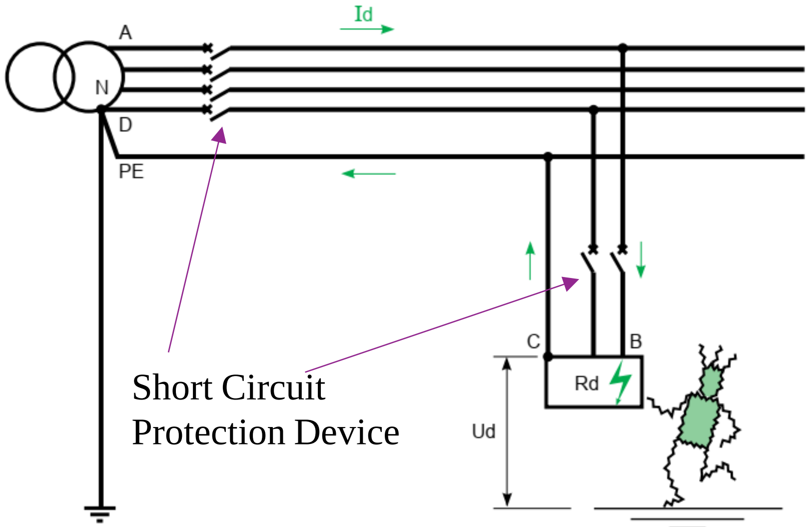
Indirect contact protection in an TN earthing system



If the line is longer than Lmax, either conductor cross-section must be increased, or it must be protected using a Residual Current Device (RCD).

Uo (volts) phase/neutral voltage	Breaking time (seconds) U_L = 50 V	Breaking time (seconds) U_L = 25 V
127	0.8	0.35
230	0.4	0.2
400	0.2	0.05
> 400	0.1	0.02

$$U_d = U_f$$

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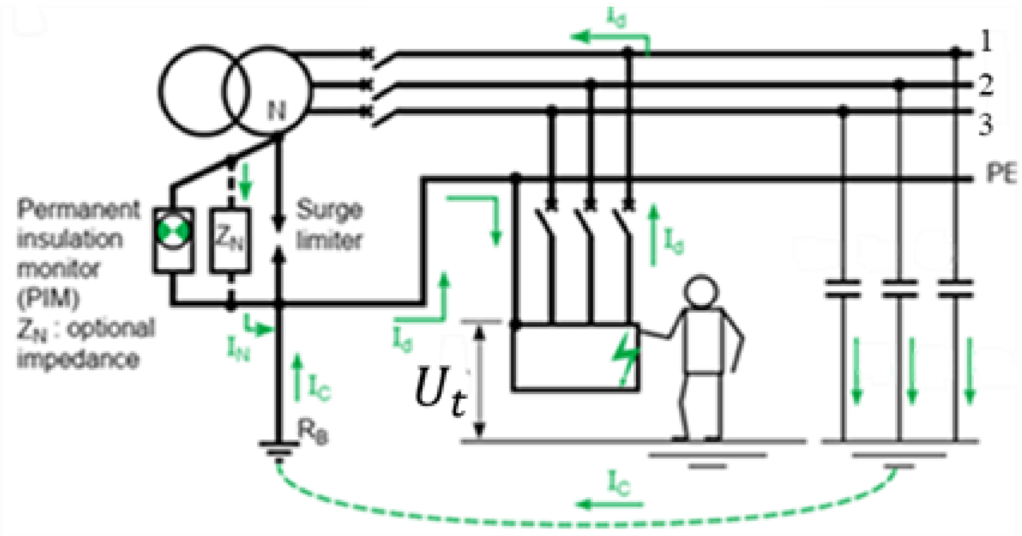
Indirect contact protection in an IT earthing system

a) Neutral earthed by impedance

- The neutral point is earthed by an impedance between 1kΩ to 2kΩ.
- The load exposed conductive parts are connected to the same earth electrode system.
- Stray capacities of the network cables create a path for the fault current.
- The stray capacitance impedance at power grid frequency is high, limiting fault current magnitude.
- The touch voltage developed in the frame-earth connection is normally reduce to a few volts. Hence is not dangerous.

First fault to earth



$$I_d = \frac{U_o}{Z_N + 3jCw} \qquad I_N = \frac{U_o}{Z_N} \qquad \overline{I_d} = \overline{I_N} + \overline{I_C}$$

$$U_t = R_B \frac{U_o}{Z_N + 3jCw} \qquad I_C = 3jCwU_0$$

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Protection and Safety Systems

Indirect contact protection in an IT earthing system

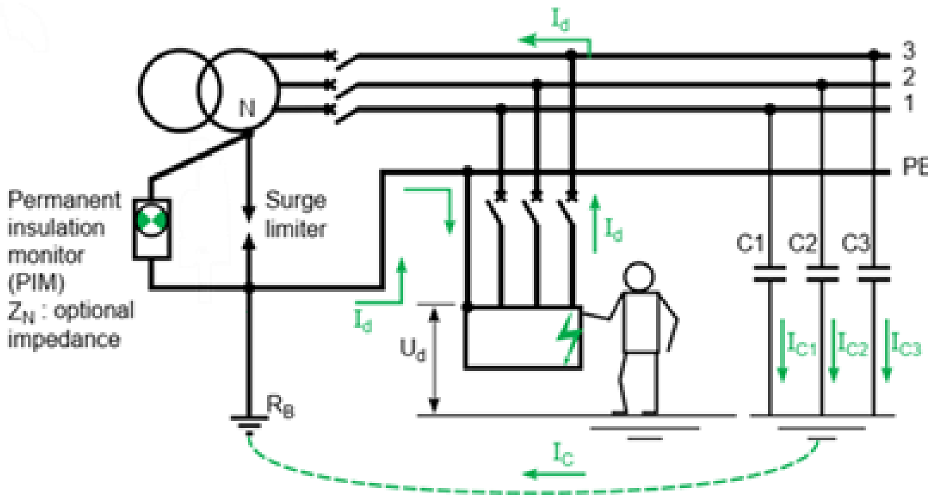
b) Unearthed neutral

- The neutral point of the power transformer secondary is not earthed.
- The load exposed conductive parts are connected to the same earth electrode system.
- The system is connected to ground through the line to ground stray capacitances.

$$\overline{I_d} = \overline{I_C} \qquad I_C = 3jCwU_0$$

$$I_d = \frac{U_o}{3jCw} \qquad U_t = R_B \frac{U_o}{3jCw}$$

First fault to earth



We assume $Z_N = \infty$.

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Indirect contact protection in an IT earthing system

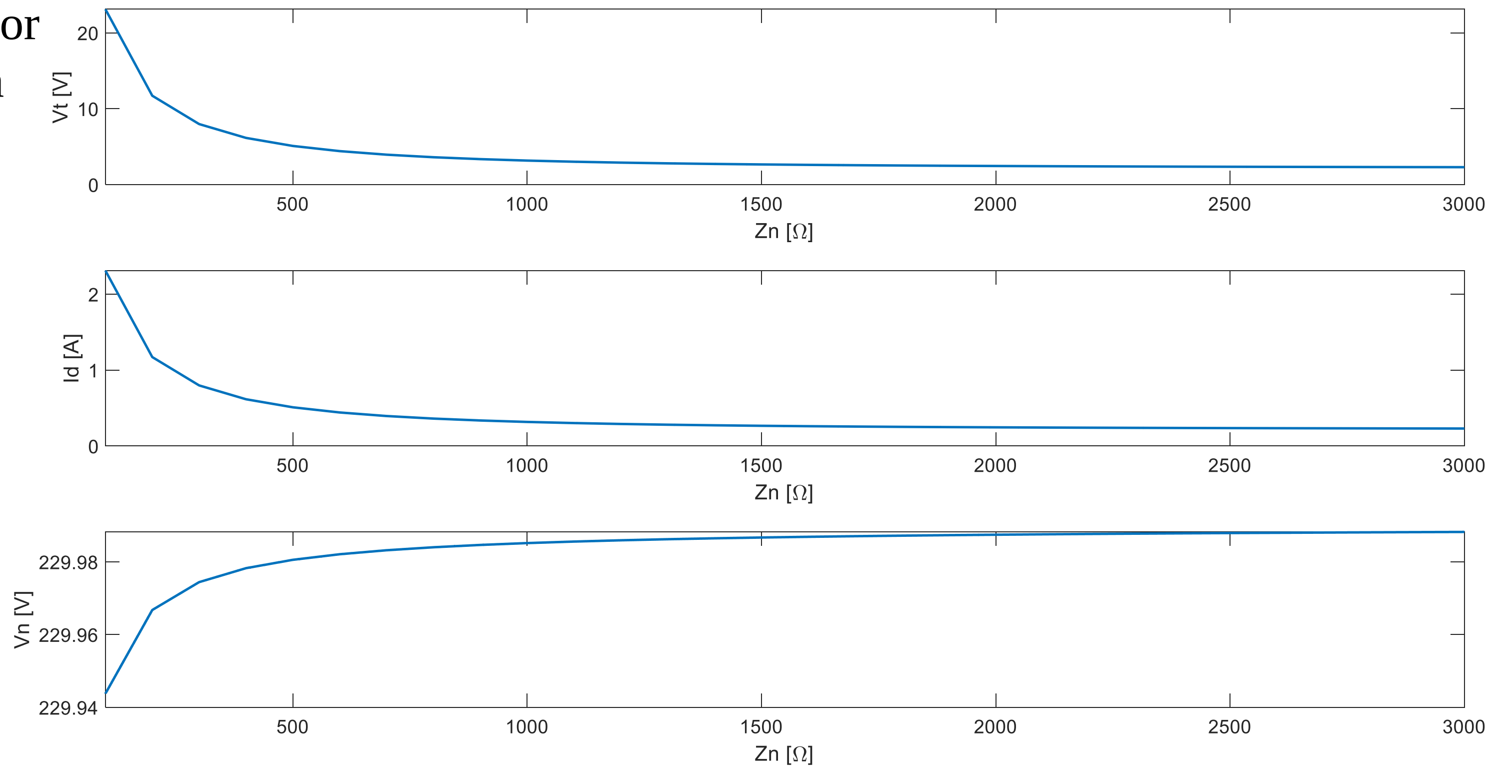
400 V_{AC} (U_o = 230), R_B = 10Ω, C=1μF

A simulation was made for an IT installation with an impedance to earth Zn variable.

A full fault (first fault) occurs on a load frame with an earth connection RB.

For a value of Z_N higher than 1000Ω, V_t is very low and constant.

Neutral voltage V_N to earth is constant.



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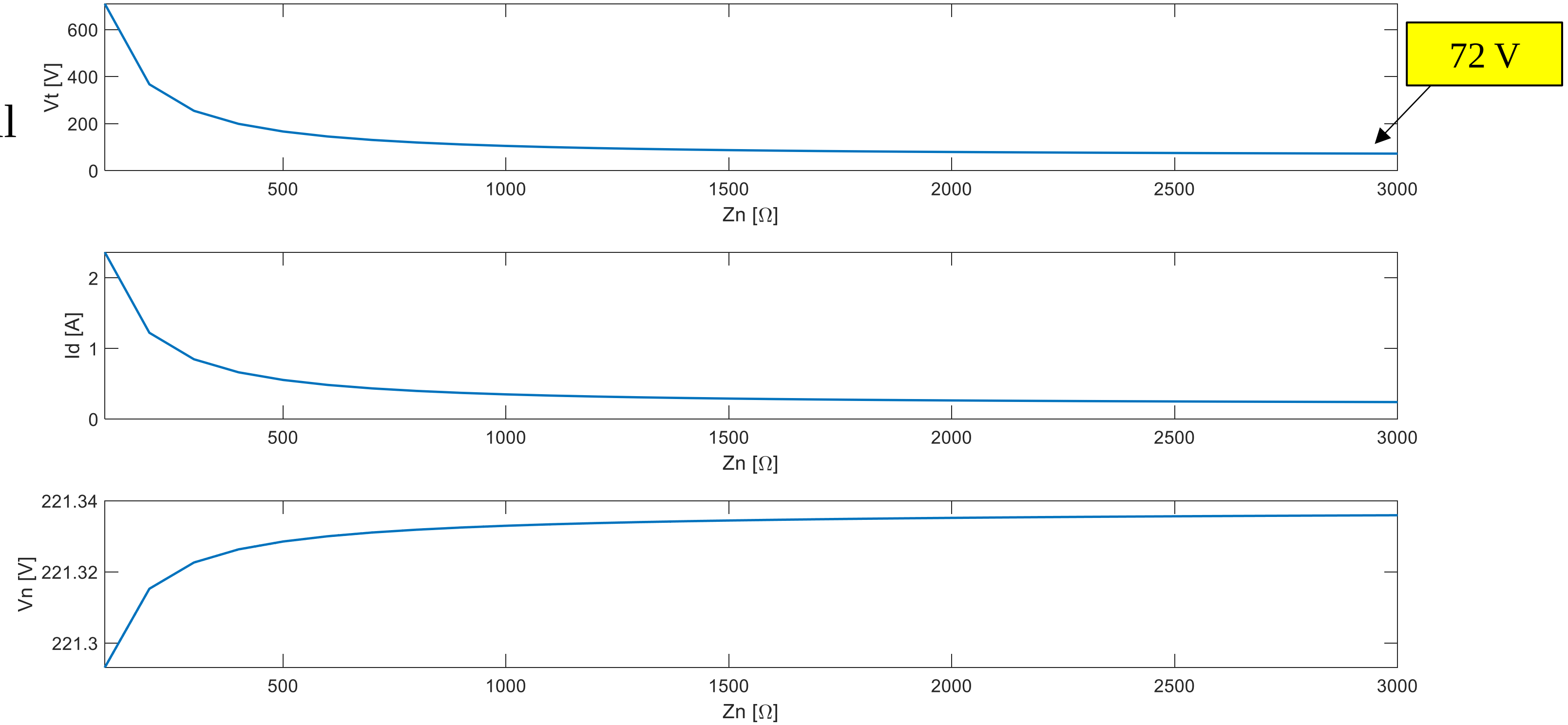
Indirect contact protection in an IT earthing system

$400\ V_{AC}$ ($U_o = 230$), $R_B = 300\Omega$, C=1μF

Touch voltage U_t stabilizes around 72 V, which it is still dangerous.

A high value of earth resistance R_B no longer guarantees that U_t is lower than 50 V, no matter if Zn is increased.



The touch voltage on a first insulation fault is always less than safety voltage, if RB does not exceed tens of ohms.

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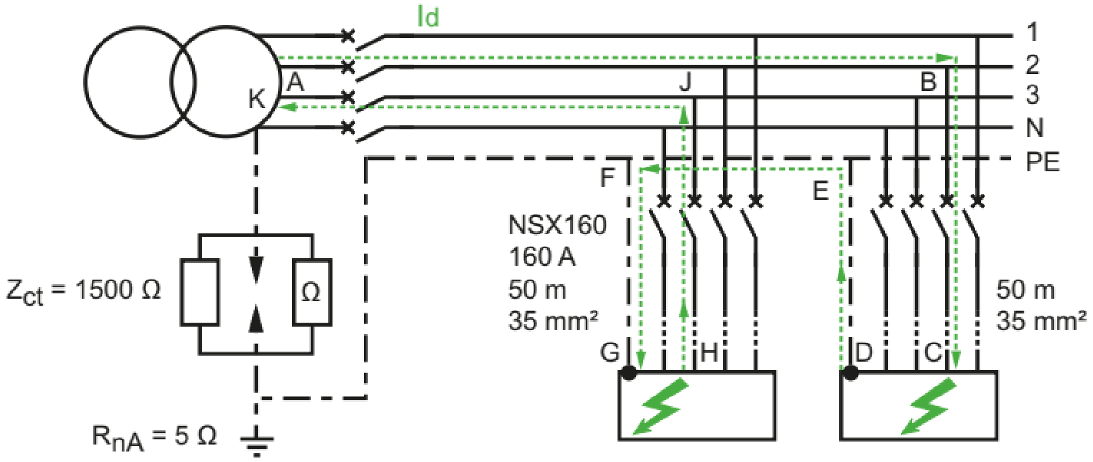
Indirect contact protection in an IT earthing system

In this case, no earth electrodes are included in the fault current path, so that a high level of fault current is assured, and conventional overcurrent protective devices are used, i.e. circuit breakers and fuses. To adjust the protective devices, the short-circuit current must be calculated, using a method applicable to TN system.



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Indirect contact protection in an IT earthing system

For the case of a 3-phase installation without neutral conductor, the second fault can generate a phase/phase short-circuit. To actuate the circuit-breaker, the voltage to apply for maximum circuit length is $\sqrt{3}U_o$ (two-line conductors close the fault path).

$$L_{max} = \frac{0.8U_0\sqrt{3}S_{ph}}{2\rho I_a(1+m)}$$

For the case of a 3-phase installation with neutral conductor, the lowest value for fault current is between the neutral conductor and line conductor. To calculate the maximum cable length only one phase to neutral voltage U_o is involved.

$$L_{max} = \frac{0.8U_0S_1}{2\rho I_a(1+m)}$$

ρ – resistivity at normal operating temperature ($23.7x10^{-3}\ mm^2/m$ for cooper, 37.6x$10^{-3}\ mm^2/m$ for aluminium)

I_a – overcurrent trip-setting level in amps

S_{ph} – cross-sectional area of the phase conductors in mm^2.

S_{PE} – cross-sectional area of PE conductor in mm^2.

S_1 – cross-sectional area of the neutral conductor in mm^2.

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Earthing system comparison

		Id	Ud	Lmax	Continuity of service
TN		$\frac{0.8\ Uo\ Sph}{\rho\ (1+m)\ L}$	$\frac{0.8\ Uo}{1+m}$	$\frac{0.8\ Uo\ Sph}{\rho\ (1+m)\ Ia}$	Vertical discrimination
TT		$\frac{Uo}{Ra\ +\ Rb}$	$\frac{Uo\ Ra}{Ra\ +\ Rb}$	No constraint	Vertical discrimination
IT	1st fault	< 1 A	$<< U_L$		No tripping
	Double fault with distributed neutral	$\leqslant \frac{1}{2}\ \frac{0.8\ Uo\ Sph}{\rho\ (1+m)\ L}$	$\leqslant \frac{m}{2}\ \frac{0.8\ Uo}{1+m}$	$\frac{1}{2}\ \frac{0.8\ Uo\ Sph}{\rho\ (1+m)\ Ia}$	Vertical discrimination and possibility of horizontal discrimination to the advantage of hich current feeders
	Double fault with non distributed neutral	$\leqslant \frac{\sqrt{3}}{2}\ \frac{0.8\ Uo\ Sph}{\rho\ (1+m)\ L}$	$\leqslant \frac{m\ \sqrt{3}}{2}\ \frac{0.8\ Uo}{1+m}$	$\frac{\sqrt{3}}{2}\ \frac{0.8\ Uo\ Sph}{\rho\ (1+m)\ Ia}$	

Uo (V AC)		50 < Uo ⩽ 120	120 < Uo ⩽ 230	230 < Uo ⩽ 400	Uo > 400
System	TN	0.8	0.4	0.2	0.1
	TT	0.3	0.2	0.07	0.04

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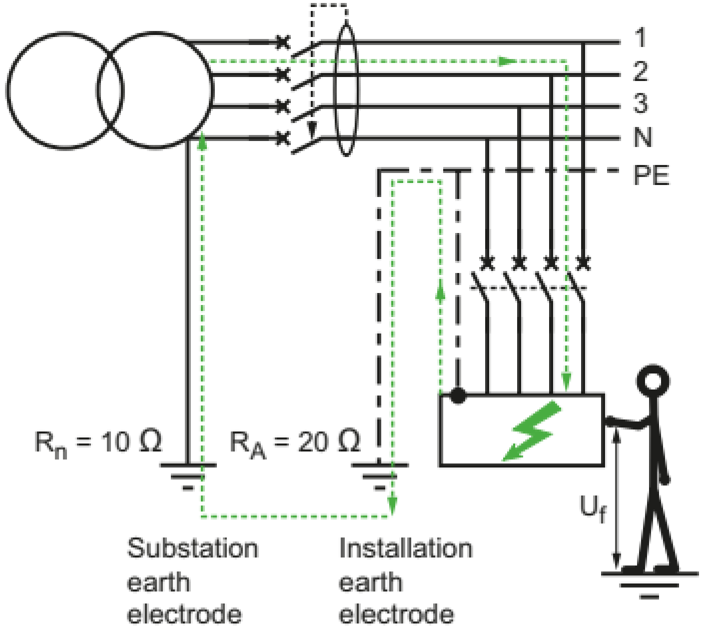
TT system example

$$I_d = \frac{230}{10+20} = 7.7\ A$$ Earth fault loop current

$$U_f = R_A I_f = 154\ V$$ Voltage is higher than 50 V

Protection by automatic disconnection is guaranteed by inserting in the circuit an RCD.



RCD rated sensitivity $I_{\Delta n}$ is given by $I_{\Delta n} \leq \frac{50}{R_A}$ $\quad I_{\Delta n} \leq 2.5\ A$

$I_{\Delta n}$	Maximum resistance of the earth electrode	
	(50 V)	(25 V)
3 A	16 Ω	8 Ω
1 A	50 Ω	25 Ω
500 mA	100 Ω	50 Ω
300 mA	166 Ω	83 Ω
30 mA	1666 Ω	833 Ω

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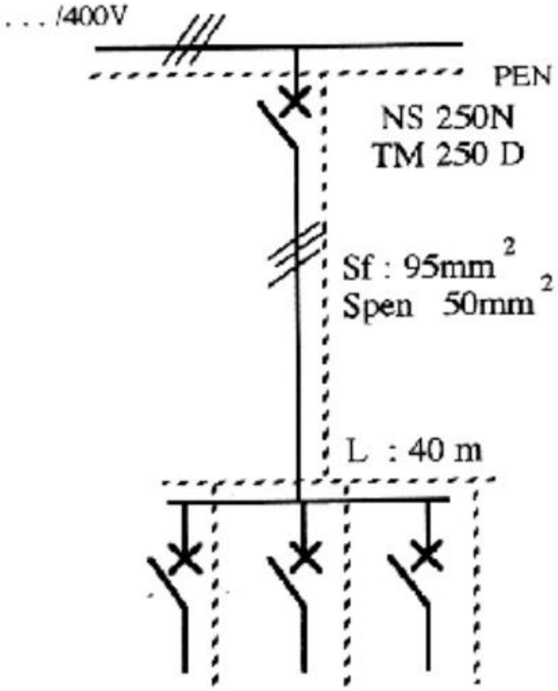
Protection and Safety Systems

TN-C example



The circuit is 40 metres long, the cross-section of the phase conductor is 95mm2 and the protective conductor is 50mm2.

The circuit is protected with an NS 250N circuit breaker (Merlin Gerin manufacturer) equipped with a TM 250 D-curve magnetothermal trip.

Check that the conditions for protection against indirect contact are guaranteed in this LV installation where the TN neutral regime is adopted.

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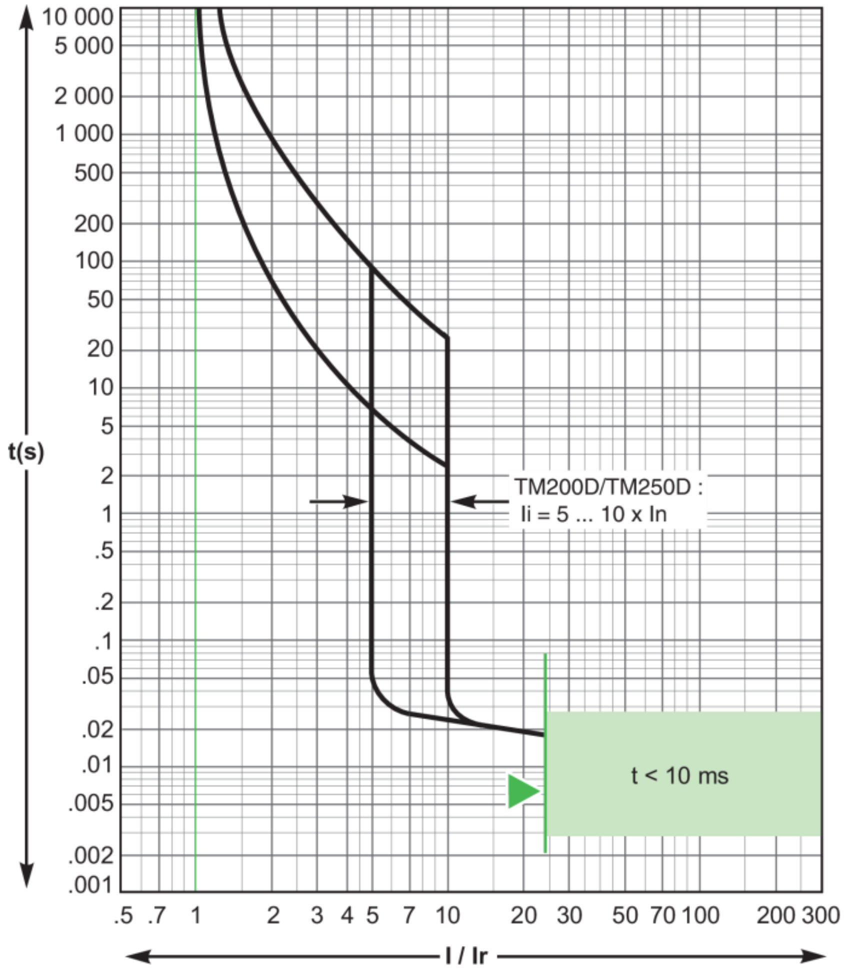
TN-C example

A basic requirement for the correct sizing of the protection device is to analyse tripping curves to get the value of the current corresponding to the operating threshold of the protection device's magnetic trigger.

The actuation range of the magnetic trip of this circuit breaker can be set between 5 and 10 times the rated current In, i.e. between 1250 and 2500A.

TM 250 D magnetothermal trip curve



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TN-C example

Fault loop impedance is given by $Z_f = \frac{0.8U_o}{I_f}$ U_o – voltage to neutral I_f – fault current

For the circuit-breaker to also guarantee protection against indirect contact, the following condition must be met $Z_f \leq \frac{0.8U_o}{I_a}$ I_a is the tripping current of the magnetic trigger of the circuit-breaker

The maximum protected length of the circuit depends on trip-current setting I_a

$$L_{ciruito} \leq \frac{0.8U_0S_{ph}}{\rho I_a(1+m)I_a}$$

$$U_f = R_{PE}\frac{0.8U_0S_{ph}}{\rho l(1+m)} \qquad R_{PE} = \rho\frac{l}{S_{PE}}$$

The maximum protected length of the circuit depends on trip-current setting I_r

Protection and Safety Systems

TN-C example

2 step verification

If the circuit breaker's magnetic trip is set to I_a=5xI_n=1250 A.

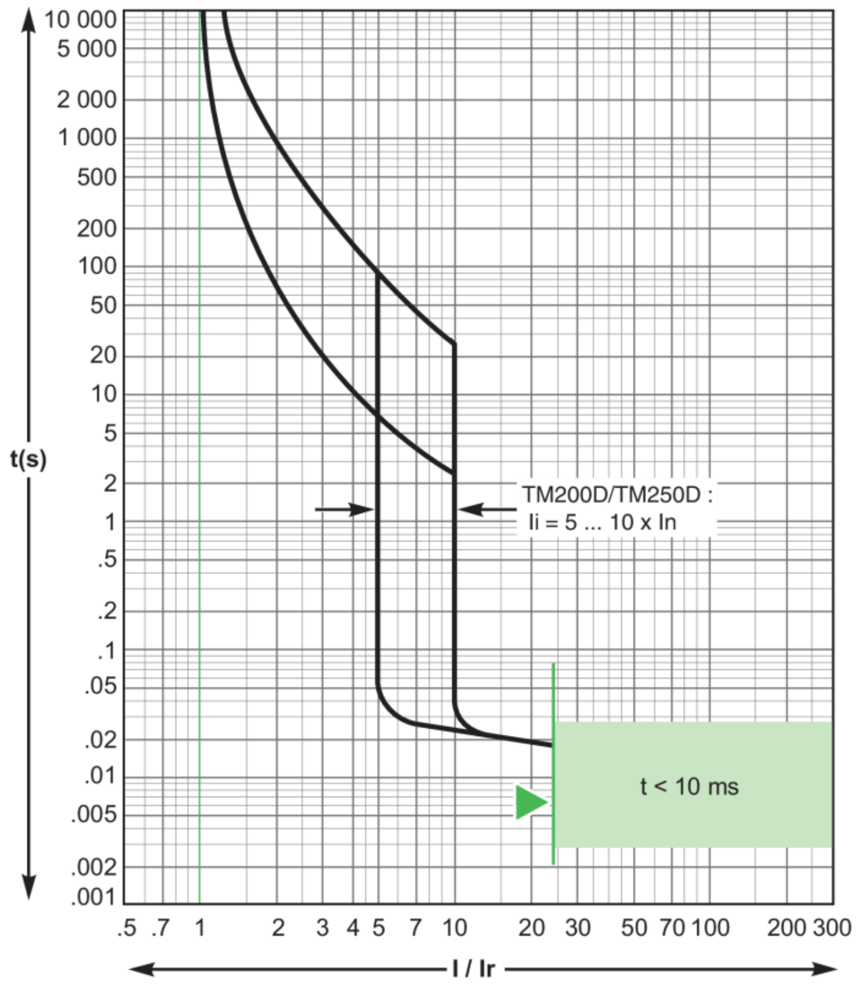
$$l_{max} \leq \frac{0.8 * 230 * 95}{0.0237 * (1 + 1.9) * 1250} \leq 203\ m$$

Now considering the circuit breaker's magnetic trip is set to I_a=10xI_n=250 A.

$$l_{max} \leq \frac{0.8 * 230 * 95}{0.0237 * (1 + 1.9) * 2500} \leq 101\ m$$

For a circuit length of 40 m both settings will provide full protection against insulation failure based short-circuit.

TM 250 D magnetothermal trip curve



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Protection and Safety Systems

TN-C example

2 step verification

The touch voltage in the load exposed conductive part for this fault is given by

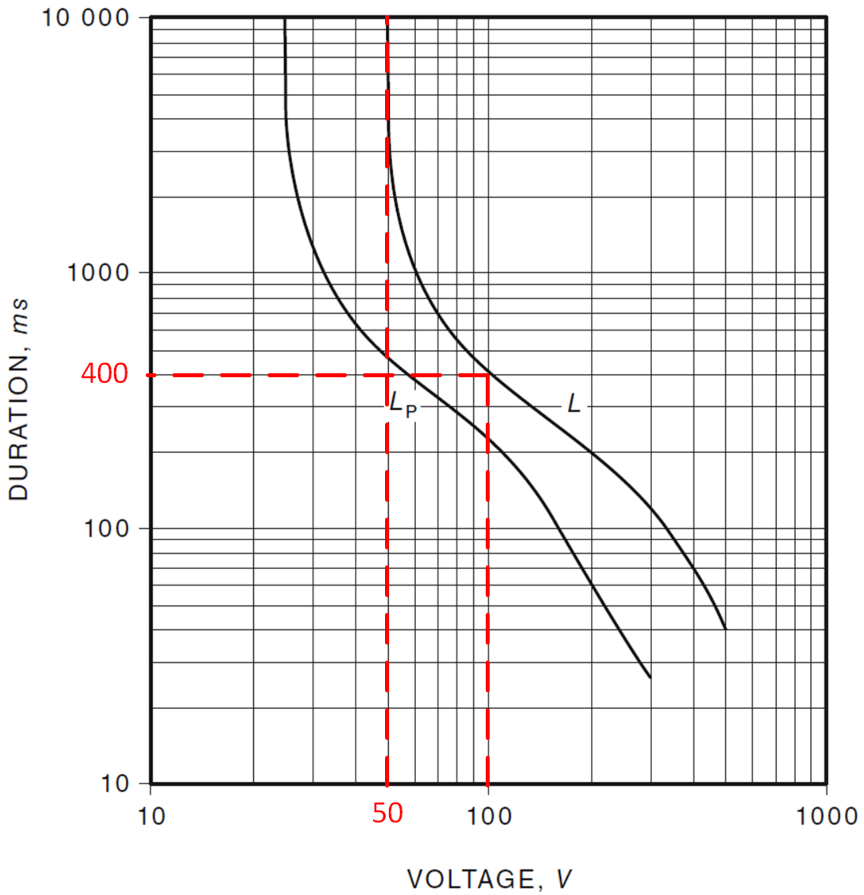
$$U_f = R_{PE}\frac{0.8 * U_o S_{ph}}{\rho * l * (1+m)} \qquad U_f = 120{,}6\ \mathrm{V}$$

Check if circuit-breaker tripping time complies with safety voltage level.

According to the standard for this voltage magnate of 120 V, disconnection time should not exceed 200 ms.

Going back to the circuit-breaker tripping curve, it is clear for a short circuit event the unit will trip in less than 50ms.

IEC 60479 standard



L for normal conditions touch voltage limit 50 V
L_p for wet conditions touch voltage limit 25 V

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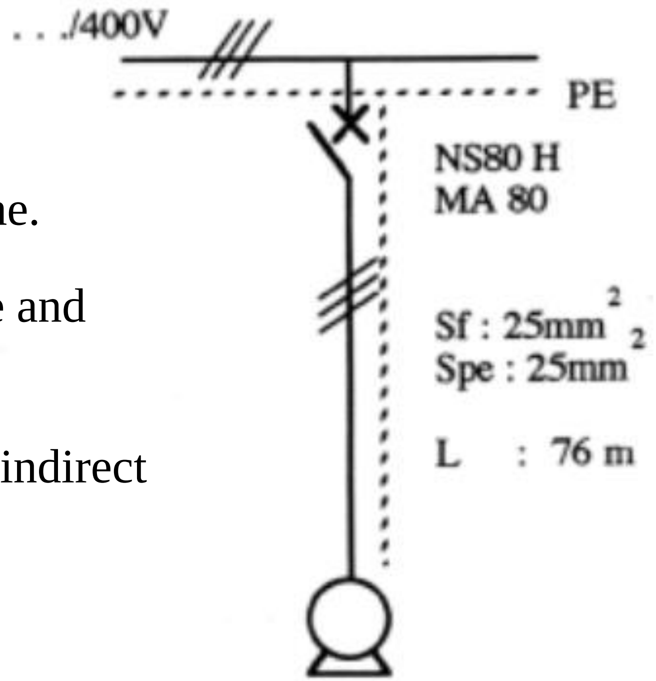
IT example without neutral distributed

The conditions for eliminating the current from a second fault are then guaranteed by the same conditions indicated for the TN scheme.

The circuit has a length of 76 metres, the cross-section of the phase and protective conductor is 25 mm^2.

Check that in this neutral regime, the protection of persons against indirect contact is effectively guaranteed with this protective device.



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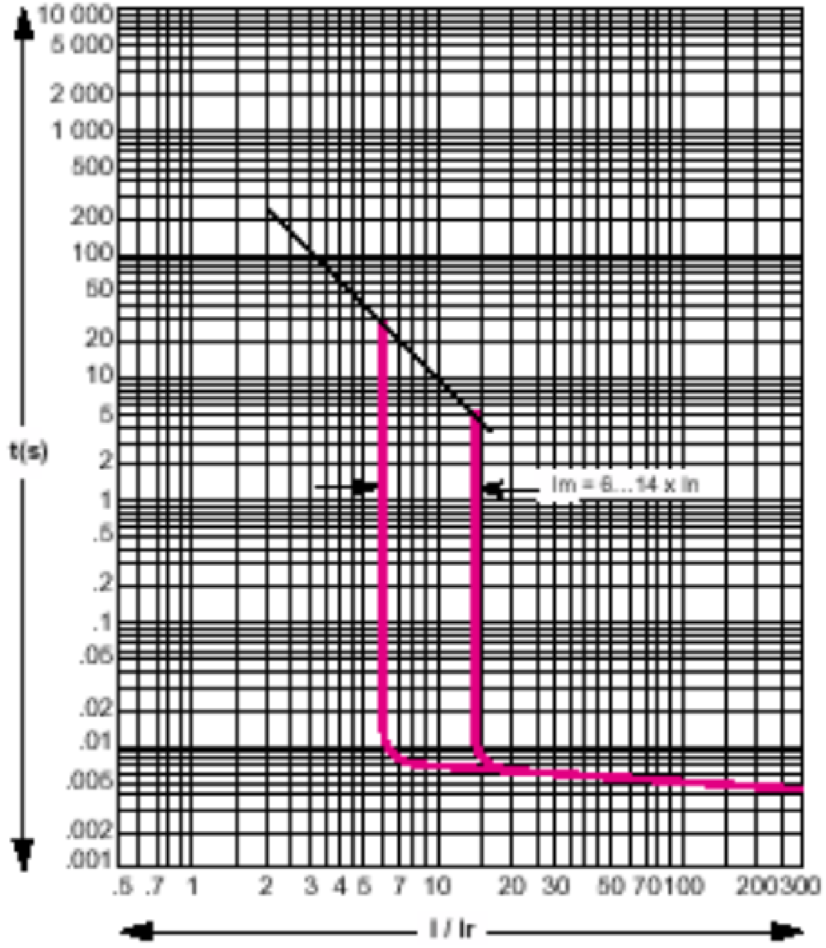
IT example without distributed neutral

It is necessary to study the device's actuation curve to obtain the value of the current corresponding to the operating threshold of the magnetic trip of the protection device.

As you can see, the actuation of the magnetic trip unit of this circuit-breaker is between 6 and 14 times the nominal value (In=80A), i.e. between 480 and 1120 A.

NS 80H circuit-breaker tripping curve



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IT example without distributed neutral

Calculation form to be used in a scenario where there is a second insulation fault (TN scheme):

$$Z_f \leq \frac{0.8\sqrt{3}U_o}{I_a}$$

Circuit fault loop impedance

Maxim fault loop impedance that CB can clear short-circuit

$$L_{ciruito} \leq \underbrace{\frac{0.8U_0\sqrt{3}S_{ph}}{2\rho I_a(1+m)I_a}}_{L_{max} \text{ protected by the circuit breaker}}$$

$$U_f = \frac{0.8 * \sqrt{3} * U_o * m}{2(1+m)}$$

$$U_f = R_{PE}I_f$$

In this neutral regime, it is considered a good approximation that at the second fault, the length of the fault loop is double that of the first fault.

CB – circuit-breaker

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NS 80H circuit-breaker tripping curve

IT example without distributed neutral

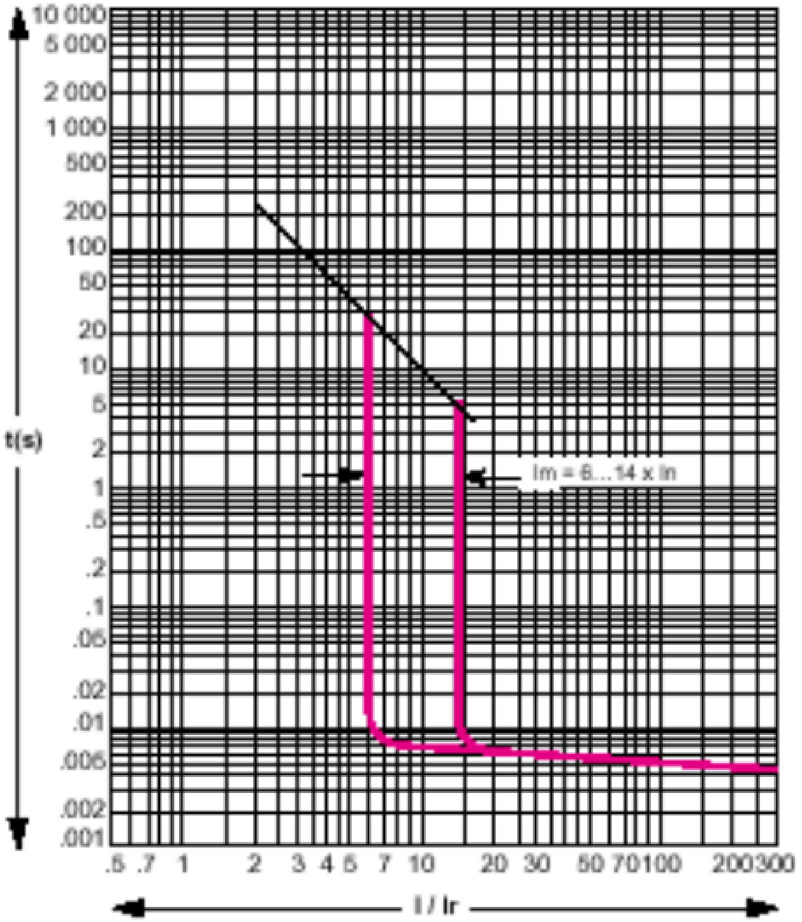
2 step verification

Set I_a=6xI_n=480 A

$$l_{max} \leq \frac{0.8 * \sqrt{3} * 230 * 25}{2 * 0.0237 * (1 + 1) * 480} \leq 175\ m$$

Set I_a=11xI_n=1120 A

$$l_{max} \leq \frac{0.8 * \sqrt{3} * 230 * 25}{2 * 0.0237 * (1 + 1) * 1120} \leq 75\ m$$



Given that the length of the circuit is 76 m, for any setting of the MA trigger (6 to 14xIn), the circuit-breaker guarantees the protection against indirect contact.

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Protection and Safety Systems

IT example without distributed neutral

2 step verification

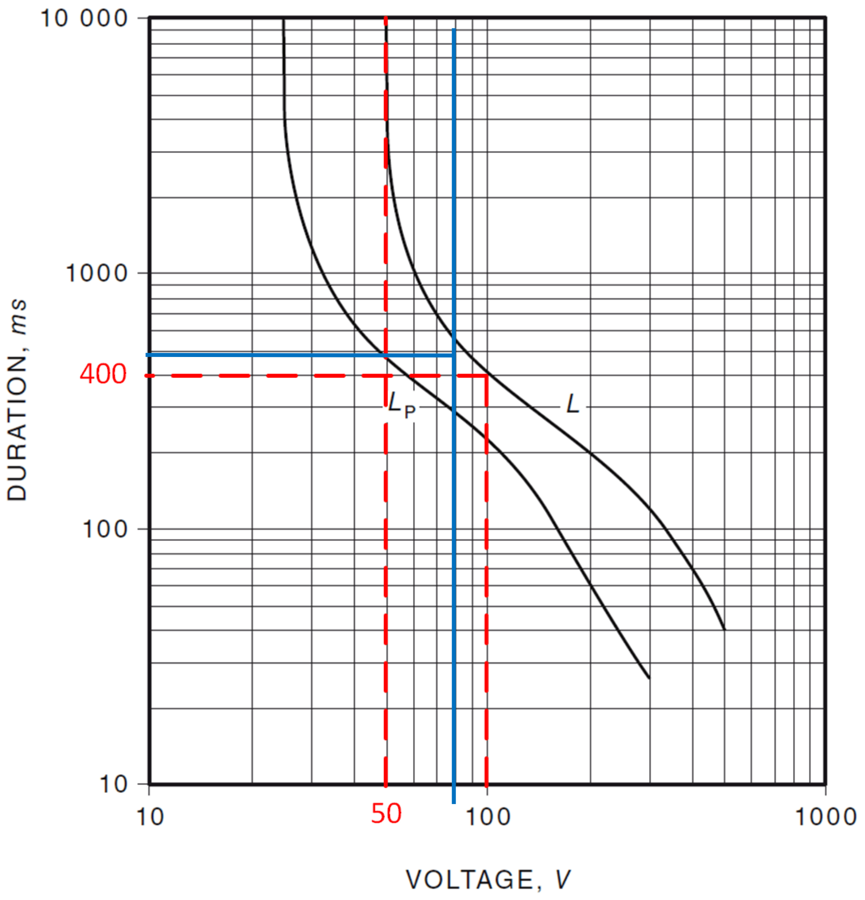
The touch voltage in case in second fault:

$U_f = 79{,}7\ \text{V}$ (Check blue trace intersection.)

To protect the person the disconnection time is below 500 ms.

Inspecting circuit-breaker tripping curve, the short circuit is cleared is less than 20ms.

IEC 60479 standard



L	for normal conditions	touch voltage limit 50 V
L_p	for wet conditions	touch voltage limit 25 V